

TASK 3 REPORT: NETWORKING, RECONNAISSANCE & SERVICE ENUMERATION

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GITHUB REPOSITORY: https://github.com/ishaliqbal/NovaShyld-Task_3

TARGET IP: 45.33.32.156 (SCANME.NMAP.ORG)

1. Introduction & Objective

In ethical hacking and penetration testing, reconnaissance is the very first and most important step. Before trying to find security flaws or weaknesses in a system, an ethical hacker needs to know what is running on that network.

The main goal of this task was to perform active network scanning on a target system using Nmap to:

- Discover active live hosts and open ports.
- Identify the exact software and service versions running on those open ports.
- Detect the underlying Operating System (OS) of the target machine.

2. Command Used & Explanation

To perform a fast, efficient, and detailed scan without overloading the network, the following command was executed in the Kali Linux terminal:

```
sudo nmap -sV -O -F 45.33.32.156
```

Breakdown of

the Command Flags:

- **sudo**: Runs the command with administrator (root) privileges, which is required for advanced OS detection and packet crafting.
- **nmap**: The core network mapping and scanning tool used for security auditing.
- **-sV**: Service Version detection flag. It probes open ports to find out the exact name and version of the software running behind them.
- **-O**: Operating System detection flag. It analyzes the target's network response behavior (TCP/IP stack fingerprinting) to guess the operating system.
- **-F**: Fast scan mode. Instead of scanning all 65,535 ports, it limits the scan to the top 100 most common ports to save time.
- **45.33.32.156**: The direct target IP address (scanme.nmap.org).

3. Scan Results & Observations

The scan completed successfully in roughly 30 seconds and provided clear details about the target's security posture.

A. Port Status Summary • Total Scanned: Top 100 ports

- **Closed Ports:** 94 ports were closed (meaning no services are running on them, so they are secure).
- **Filtered Ports:** 4 ports (13, 79, 389, and 444) were marked as **filtered**, indicating that a firewall or packet filter is actively blocking or dropping traffic on those ports.
- **Open Ports:** 2 key ports were found completely open and active.

B. Detailed Service & Version Table

Port Number	Protoc	State	Service Name	Version & Details
22	TCP	Open	SSH	OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux)
80	TCP	Open	HTTP	Apache httpd 2.4.7 ((Ubuntu))
13	TCP	Filtered	daytime	Firewall protected / No direct response
79	TCP	Filtered	finger	Firewall protected / No direct response
389	TCP	Filtered	ldap	Firewall protected / No direct response

444	TCP		Filtered	snpp	Firewall protected / No direct response
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4. Operating System (OS) Fingerprinting

Using the -O flag, Nmap analyzed how the target system handles network packets.

- **Detected OS Family:** Linux
- **Kernel Information:** Linux Kernel environment (cpe:/o:linux:linux_kernel)
- **Target Environment:** The target is running on a standard Ubuntu-based Linux configuration.

5. Conclusion & Security Takeaways

1. **Understanding the Attack Surface:** The scan revealed that the target machine exposes only a minimal attack surface—specifically, an SSH server for remote management (Port 22) and an Apache web server for hosting web content (Port 80).
2. **Version Disclosure Risk:** Service version scanning (-sV) successfully identified older software releases (OpenSSH 6.6.1p1 and Apache 2.4.7). In real-world security

assessments, running outdated software versions can give attackers clues about known software bugs or patches that haven't been applied yet.

3. **Firewall Effectiveness:** The presence of filtered ports shows that network boundaries and firewalls are active, preventing unauthorized probes on nonessential ports.

Here the evidence & screenshot:

